

Recommendation Report (Part D)

Which model is best between Model 1 and Model 2?

Between the two models XGBoost performed better overall, although the difference was minimal. It achieved the highest average precision (0.472 vs 0.465) and an ROC_AUC (0.749 vs 0.748), as I said the differences are minimal.

When computing the brier scores, XGBoost scores lower which indicates a stronger probability accuracy.

The two models are essentially identical in performance where the main difference comes in is with computation efficiency. As the data will scale (with the inclusion of the viewing history and other datasets) the complexity will also increase and having a model that can still perform well with less time and resources spent on compute the better.

My recommendation is therefore the XGBoost Model.

How can the selected model be improved?

The current model is good at predicting based on global trends but not on local trends. This is mostly due to the missing integration of the platform session data which is a dataset that shows from the platform titles how many are watched. In my original scope this piece of data was supposed to identify feed into the modelling to help the model understand what is currently being watched.

I did generate this data but never built on it due to file size and compute concerns. I think this is a crucial growth point and will significantly improve the models effectiveness, similarly to the takeaways from Publication 2 where the inclusion of contextual data improve the model significantly.

There is a similar growth point with the search query data that wasn't used as a proxy for demand that would have given the model extra context. Although search demand in the publication 3 of the literature review did not necessarily improve the baseline I think in this case it will as it will localise the demand which will help the model predict watchability better.

Next the model would benefit from the inclusion of more metadata points, something I hope to achieve by tweaking the TMDB API calls, those metadata fields can include: actors, directors, language, country of origin, runtime, studio / distributor, original release recency. content type

How will the selected model need to be adapted to work with your requested data from SS1?

As mentioned above right now the model mostly learns the relationships between genres and uses synthetic data to make a prediction. Using the requested data there will be a true outcome (ie something watched vs my simulated outcome). It will also not only rely solely on metadata about the titles themselves but also ingest historic viewing history into the

equation.

The XGBoost model structure would remain the same but as mentioned above the data would change. Instead of basing the outcome of 150 household views on synthetic data we would be able to base that on actual viewing data.

Does your results align with the related literature from Part A?

One of the key takeaways from publication 2 was that “tree-based gradient boosting models handle this type of data well” and this seems to hold for the data at hand.